GCP DEVOPS INTERVIEW QUESTIONS AND ANSWERS

Target Role: DevOps Engineer

Focus: Google Cloud Platform (GCP)

Experience Context: Compute Engine, VPC, IAM, Cloud Storage, Cloud Load Balancing, Managed Instance Groups, Cloud Monitoring, GKE, Terraform, Jenkins, Docker and Kubernetes

1. WHAT IS GCP?

Answer:

GCP stands for Google Cloud Platform. It is Google's cloud computing platform that provides services for compute, storage, networking, security, databases, containers, monitoring and application deployment.

As a DevOps Engineer, I used GCP services to provision infrastructure, deploy applications, configure networking, manage access and monitor resources.

2. WHICH GCP SERVICES HAVE YOU WORKED WITH?

Answer:

I have worked with:

- Compute Engine
- VPC
- IAM
- Cloud Storage
- Cloud Load Balancing
- Managed Instance Groups
- Cloud Monitoring
- Google Kubernetes Engine (GKE)

I also used Terraform to provision and manage GCP infrastructure.

3. WHAT IS COMPUTE ENGINE?

Answer:

Compute Engine is GCP's Infrastructure-as-a-Service offering that provides virtual machines.

We can configure:

- Machine type
- CPU
- Memory
- Boot disk
- Operating system
- Network
- Firewall rules
- Service account

I used Compute Engine for hosting application and infrastructure workloads.

4. HOW DO YOU CREATE A VM IN GCP?

Answer:

A VM can be created through the GCP Console, gcloud CLI or Terraform.

Using gcloud:

```
gcloud compute instances create my-vm
--zone=us-central1-a
--machine-type=e2-medium
```

After creating the VM, I would verify:

- VM status
- Network configuration
- Firewall rules
- Service account
- SSH connectivity
- Application/service status

5. WHAT IS VPC IN GCP?

Answer:

VPC stands for Virtual Private Cloud.

A VPC provides networking for GCP resources. It allows us to configure:

- Subnets

- Routes
- Firewall rules
- IP addresses
- Network connectivity

Example:

VPC | +-- Subnet | +-- VM | +-- GKE

6. WHAT IS A SUBNET?

Answer:

A subnet is a logical IP address range within a VPC network.

For example:

VPC | +-- us-central1 subnet | +-- us-east1 subnet

The subnet defines the IP address range in which resources such as VMs can receive IP addresses.

7. WHAT IS THE DIFFERENCE BETWEEN VPC AND SUBNET?

Answer:

VPC is the overall virtual network.

Subnet is a smaller network segment inside the VPC.

Example:

VPC | +-- Subnet 1 | +-- Subnet 2 | +-- Subnet 3

8. WHAT IS A FIREWALL RULE IN GCP?

Answer:

A GCP VPC firewall rule controls network traffic to and from supported resources.

Rules can control:

- Ingress traffic
- Egress traffic
- Source ranges
- Destination ranges
- Protocols
- Ports

For example, if an application runs on port 8080, the appropriate firewall configuration must allow required traffic to that port.

9. WHAT IS IAM IN GCP?

Answer:

IAM stands for Identity and Access Management.

IAM controls:

WHO can access a resource WHAT they can do WHICH resource they can access

The basic model is:

Principal + Role + Resource

For example:

Jenkins Service Account | v Required GCP Role | v GCP Resource

10. WHAT IS A SERVICE ACCOUNT?

Answer:

A service account is an identity used by applications, VMs, Jenkins pipelines or other workloads to authenticate with GCP.

For example, Jenkins can use a service account with appropriate IAM permissions to access GCP resources.

11. USER ACCOUNT VS SERVICE ACCOUNT

Answer:

User Account:

- Used by humans.
- Usually represents an individual user.
- Used for interactive access.

Service Account:

- Used by applications and workloads.
- Used for programmatic authentication.
- Can be assigned IAM roles.

Example:

Developer -> User Account Jenkins -> Service Account

12. WHAT IS THE PRINCIPLE OF LEAST PRIVILEGE?

Answer:

The principle of least privilege means giving a user or service only the permissions required to perform its task.

For example, if Jenkins only needs permission to push images to a specific repository, we should avoid giving it unnecessary administrator-level permissions.

13. WHAT IS CLOUD STORAGE?

Answer:

Cloud Storage is GCP's object storage service.

It can be used to store:

- Files
- Backups
- Logs
- Build artifacts
- Static content
- Application data

Cloud Storage organizes data using buckets and objects.

14. WHAT IS A BUCKET?

Answer:

A bucket is a container used to store objects in Cloud Storage.

Example:

Bucket | +-- application.zip +-- backup.tar.gz +-- logs/ | +-- application.log

15. WHAT IS CLOUD LOAD BALANCING?

Answer:

Cloud Load Balancing distributes incoming network traffic across backend resources.

Example:

Users | v Load Balancer | +---- VM1 | +---- VM2 | +---- VM3

It helps distribute traffic and improve application availability.

16. WHAT IS A HEALTH CHECK?

Answer:

A health check verifies whether a backend resource is healthy and able to receive traffic.

For example, a load balancer can periodically check an application endpoint or port.

If a backend fails the configured health check, traffic can be directed to healthy backends.

17. WHAT IS A MANAGED INSTANCE GROUP?

Answer:

A Managed Instance Group, or MIG, is a collection of VM instances managed as a group.

MIG provides capabilities such as:

- Autoscaling
- Autohealing

- Rolling updates
- Instance management

Example:

Load Balancer | v MIG / |
VM1 VM2 VM3

18. WHAT IS AUTOSCALING?

Answer:

Autoscaling automatically adjusts the number of VM instances based on configured conditions.

For example:

High traffic | v CPU utilization increases | v Autoscaler adds VM instances | v Traffic is distributed

19. WHAT IS AUTOHEALING?

Answer:

Autohealing helps maintain the health of instances in a Managed Instance Group.

If an instance becomes unhealthy according to the configured health check, the MIG can recreate the instance.

20. WHAT IS THE DIFFERENCE BETWEEN AUTOSCALING AND AUTOHEALING?

Answer:

Autoscaling changes the number of instances based on workload.

Autohealing replaces unhealthy instances.

Autoscaling: High load -> Add/Remove instances

Autohealing: Unhealthy instance -> Recreate instance

21. WHAT IS CLOUD MONITORING?

Answer:

Cloud Monitoring is used to monitor GCP resources and applications.

It provides metrics and monitoring capabilities for resources such as VMs and other supported services.

We can use it to monitor:

- CPU utilization
- Network traffic
- Disk-related metrics
- Resource health
- Application metrics

We can also configure alerting policies.

22. HOW DO YOU TROUBLESHOOT HIGH CPU ON A GCP VM?

Answer:

I follow a systematic troubleshooting approach.

1. Check Cloud Monitoring.
2. Identify the affected VM.
3. Check CPU utilization.
4. Connect to the VM if access is available.
5. Run Linux commands:

top ps -ef uptime

1. Identify the process consuming CPU.
2. Check application logs.
3. Determine the root cause.
4. Take corrective action.
5. Monitor the VM after the fix.

23. HOW DO YOU TROUBLESHOOT HIGH DISK USAGE?

Answer:

First I check disk usage:

df -h

Then I identify large directories:

du -sh /*

I check application and log directories.

I also check whether old logs, temporary files or unnecessary files are consuming disk space.

After identifying the cause, I clean up according to the application and operational procedures or increase the disk capacity if required.

24. WHAT IS GKE?

Answer:

GKE stands for Google Kubernetes Engine.

It is Google's managed Kubernetes service used to deploy and manage containerized applications.

Typical architecture:

GKE Cluster | +-- Node | | | +-- Pod | +-- Node | +-- Pod

25. GKE VS COMPUTE ENGINE

Answer:

Compute Engine provides virtual machines.

GKE provides a managed Kubernetes platform for running containerized applications.

Compute Engine:

Application | v VM

GKE:

Application | v Kubernetes | v Pod | v Container

26. HOW DO YOU DEPLOY A DOCKER APPLICATION TO GKE?

Answer:

The general process is:

1. Build the Docker image.
2. Tag the image.
3. Push the image to a container registry.
4. Create or update Kubernetes manifests.
5. Deploy the application to GKE.
6. Verify Pods and Services.

Example:

Docker Image | v Container Registry | v GKE | v Kubernetes Pod

27. HOW DO YOU CHECK GKE CLUSTER STATUS?

Answer:

Using gcloud:

gcloud container clusters list

To get cluster credentials:

gcloud container clusters get-credentials CLUSTER_NAME
--region REGION

Then:

kubectl get nodes kubectl get pods

28. A GKE POD IS NOT STARTING. HOW DO YOU TROUBLESHOOT IT?

Answer:

First:

kubectl get pods

Then:

kubectl describe pod <pod-name>

and:

kubectl logs <pod-name>

I check:

- Pod events
- Container logs
- Image name
- Image tag
- Registry access
- Resource limits
- ConfigMaps
- Secrets
- Environment variables
- Liveness/readiness probes

29. WHAT IS GOOGLE CLOUD CLI?

Answer:

Google Cloud CLI, commonly called gcloud CLI, is a command-line tool used to manage GCP resources.

Examples:

gcloud auth login gcloud projects list gcloud compute instances list gcloud container clusters list

30. HOW DO YOU CHECK THE CURRENT GCP PROJECT?

Answer:

I can use:

gcloud config get-value project

To change the project:

gcloud config set project PROJECT_ID

31. HOW DO YOU AUTHENTICATE USING GCP CLI?

Answer:

For interactive user authentication:

gcloud auth login

For application or automation scenarios, service accounts or other supported workload authentication mechanisms can be used depending on the environment.

32. WHAT IS TERRAFORM AND HOW DID YOU USE IT WITH GCP?

Answer:

Terraform is an Infrastructure as Code tool.

I used Terraform to define and manage GCP infrastructure using configuration files.

Typical workflow:

terraform init terraform validate terraform plan terraform apply

33. WHAT IS TERRAFORM PLAN?

Answer:

terraform plan shows the changes Terraform intends to make.

It allows us to review infrastructure changes before applying them.

34. WHAT IS TERRAFORM APPLY?

Answer:

terraform apply executes the Terraform configuration and applies the planned infrastructure changes.

35. WHAT IS TERRAFORM STATE?

Answer:

Terraform state keeps information about the infrastructure resources managed by Terraform.

Terraform uses the state to determine the relationship between the configuration and the actual infrastructure.

36. HOW DO YOU TROUBLESHOOT A TERRAFORM DEPLOYMENT FAILURE IN GCP?

Answer:

I would check:

1. Terraform configuration.
2. terraform validate output.
3. terraform plan output.
4. Authentication.
5. GCP project ID.
6. Required IAM permissions.
7. Required GCP APIs.
8. Resource dependencies.
9. Quotas and limits.
10. Terraform state.

I would then correct the issue and run plan again before applying.

37. JENKINS CANNOT ACCESS A GCP RESOURCE. WHAT WILL YOU CHECK?

Answer:

I would check:

1. Jenkins credentials.
2. Service account.
3. IAM roles.
4. GCP project ID.
5. Authentication configuration.
6. Required APIs.

7. Network connectivity.
8. Jenkins pipeline configuration.
9. Jenkins console logs.

I would verify that the Jenkins identity has only the permissions required for the operation.

1. A VM IS RUNNING BUT THE APPLICATION IS NOT ACCESSIBLE. WHAT WILL YOU CHECK?
   ==========

Answer:

I would check:

1. VM status.
2. Application/service status.
3. Listening port.
4. Firewall rules.
5. VPC configuration.
6. Routes.
7. Load balancer configuration if applicable.
8. Health checks.
9. Application logs.
10. Network connectivity.

For example, on Linux:

systemctl status <service> ss -lntp curl localhost:<port>

39. LOAD BALANCER IS NOT ROUTING TRAFFIC. HOW DO YOU TROUBLESHOOT?

Answer:

I would check:

- Load balancer configuration.
- Frontend IP and port.
- Backend service.
- Backend instances.
- Health checks.
- Firewall rules.
- VPC configuration.
- Application listening port.
- Backend logs.

The key point is to determine whether the problem is at the frontend, load-balancer, backend-health, network or application layer.

40. WHAT HAPPENS IF A VM IN A MIG BECOMES UNHEALTHY?

Answer:

The configured health check detects the unhealthy instance.

If autohealing is configured, the Managed Instance Group can recreate the unhealthy VM according to its configuration.

Traffic can continue to be served by healthy instances.

41. WHAT IS A REGION IN GCP?

Answer:

A region is a geographical area containing multiple zones.

Examples include:

- us-central1
- us-east1
- europe-west1
- asia-south1

The exact regions used depend on the application's requirements.

42. WHAT IS A ZONE?

Answer:

A zone is an isolated location within a GCP region where resources such as zonal VMs can run.

Example:

Region: us-central1

Zones: us-central1-a us-central1-b us-central1-c

43. REGION VS ZONE

Answer:

Region: A geographical area containing multiple zones.

Zone: A specific location within a region where zonal resources can run.

Example:

us-central1 | +-- us-central1-a +-- us-central1-b +-- us-central1-c

44. WHAT IS HIGH AVAILABILITY IN GCP?

Answer:

High availability means designing infrastructure so that an application can continue operating when individual resources fail.

Depending on the architecture, this can involve:

- Multiple VM instances
- Managed Instance Groups
- Load Balancing
- Multiple zones
- Health checks
- Autohealing
- Appropriate application architecture

45. WHAT IS A STATIC IP ADDRESS?

Answer:

A static external IP address is a reserved IP address that remains associated with a resource until it is released or reassigned.

It can be useful when a service needs a stable IP address.

46. WHAT IS THE DIFFERENCE BETWEEN INTERNAL AND EXTERNAL IP?

Answer:

Internal IP: Used for communication within the VPC and connected networks.

External IP: Used for communication with resources over the public internet, subject to the configured network and security controls.

47. WHAT IS CLOUD NAT?

Answer:

Cloud NAT provides network address translation for resources that have private IP addresses and need outbound internet connectivity.

It allows private resources to initiate outbound connections without requiring an external IP on each resource.

48. WHAT IS CLOUD DNS?

Answer:

Cloud DNS is Google's managed DNS service.

It can be used to manage DNS zones and records such as:

- A records
- CNAME records
- MX records
- TXT records

49. HOW WOULD YOU SECURE A GCP ENVIRONMENT?

Answer:

I would follow a layered security approach:

- Use IAM with least privilege.
- Use service accounts for workloads.
- Restrict firewall rules.
- Use private networking where appropriate.
- Avoid unnecessary public IP exposure.
- Use encryption.
- Monitor activity and resources.
- Rotate credentials where applicable.

• Apply organization/project policies where applicable.
• Keep systems and applications patched.

50. EXPLAIN YOUR GCP PROJECT ARCHITECTURE.

Answer:

"In my project, GCP was used as the cloud platform for hosting and supporting application infrastructure.

Terraform was used for Infrastructure as Code. We provisioned resources such as VPC, Compute Engine and other required GCP components.

IAM was used to control access to resources. Managed Instance Groups and Cloud Load Balancing were used where required for scalable and highly available VM-based workloads.

Cloud Storage was used for object storage, and Cloud Monitoring was used for monitoring infrastructure and configuring alerts.

For containerized applications, we used GKE. Jenkins was used as the CI/CD tool. The pipeline built the application, created Docker images, pushed them to the registry and deployed them to Kubernetes.

So the overall flow was:

Git | v Jenkins | v Build | v Docker Image | v Container Registry | v GKE | v Application

Infrastructure was managed using:

Terraform | v GCP | +-- VPC +-- Compute +-- IAM +-- Load Balancing +-- Managed Instance Groups +-- Monitoring"

51. PRODUCTION SCENARIO – GCP VM IS DOWN

Answer:

"If I receive an alert that a production VM is down, first I verify the alert in Cloud Monitoring and check the VM status.

Then I check whether the VM is reachable and whether the issue is related to the VM, network, application or load balancer.

If it is part of a Managed Instance Group, I check the MIG status and health checks.

I review relevant logs and metrics, identify the root cause and restore service according to the production support procedure.

After restoration, I verify application availability and document the incident and root cause."

52. PRODUCTION SCENARIO – HIGH CPU

Answer:

"If Cloud Monitoring reports high CPU, I first identify the affected VM and check the CPU metric over time.

I then connect to the server and use commands such as top and ps to identify the process consuming CPU.

I check application logs and recent changes to understand the cause.

After identifying the issue, I take the appropriate corrective action and monitor the system to confirm that CPU utilization has returned to normal."

53. PRODUCTION SCENARIO – GCP PERMISSION DENIED

Answer:

"If I receive a permission-denied error, I first identify which identity is making the request.

Then I check the IAM role assigned to that identity and verify whether the required permission is included.

I also check whether the correct GCP project and resource are being used.

I follow the least-privilege approach and request or assign only the required permission."

54. IMPORTANT GCP COMMANDS

Authentication:

gcloud auth login

Current project:

gcloud config get-value project

Set project:

gcloud config set project PROJECT_ID

List projects:

gcloud projects list

List VMs:

gcloud compute instances list

List GKE clusters:

gcloud container clusters list

Get GKE credentials:

gcloud container clusters get-credentials CLUSTER_NAME
--region REGION

Kubernetes nodes:

kubectl get nodes

Kubernetes Pods:

kubectl get pods

Pod details:

kubectl describe pod <pod-name>

Pod logs:

kubectl logs <pod-name>

55. QUICK REVISION – MOST IMPORTANT QUESTIONS

1. What is GCP?
2. What is Compute Engine?
3. What is VPC?
4. What is a subnet?
5. VPC vs subnet?
6. What is a firewall rule?
7. What is IAM?
8. What is a service account?

9. User account vs service account?
10. What is Cloud Storage?
11. What is a bucket?
12. What is Cloud Load Balancing?
13. What is a health check?
14. What is a Managed Instance Group?
15. Autoscaling vs autohealing?
16. What is Cloud Monitoring?
17. What is GKE?
18. GKE vs Compute Engine?
19. How do you deploy Docker applications to GKE?
20. How do you troubleshoot a failed GKE Pod?
21. What is Terraform?
22. How do you use Terraform with GCP?
23. What is terraform plan?
24. What is terraform apply?
25. How do you troubleshoot a GCP VM?
26. How do you troubleshoot high CPU?
27. How do you troubleshoot high disk usage?
28. How do you troubleshoot a Load Balancer?
29. How do you troubleshoot IAM permission issues?
30. Explain your complete GCP project architecture.

END OF GCP DEVOPS INTERVIEW QUESTIONS